

--Question Starting--

Match the following concepts with their respective applications or phenomena:

1. Concept Phenomenon

I. Young's modulus A. Rate of change of velocity for a falling object reaching constant speed

II. Bernoulli's principle B. Amount of deformation per unit area under tensile stress

III. Stoke's law C. Describes fluid flow and change in pressure across different points

IV. Surface tension D. Rise of liquid in a narrow tube against gravity

Choose the correct answer from the options given below:

(1) I-B, II-C, III-A, IV-D

(2) I-D, II-A, III-B, IV-C

(3) I-C, II-D, III-B, IV-A

(4) I-A, II-B, III-D, IV-C

Answer Key: 1

Solution:

? Young's modulus: Measures the stiffness of a solid material and is used to calculate how much it will deform under a certain load, hence relating to B.

? Bernoulli's principle: Explains how an increase in the speed of a fluid occurs simultaneously with a decrease in pressure or potential energy, hence relating to C.

? Stoke's law: Describes the drag force experienced by spherical objects moving through a viscous fluid, which affects the terminal velocity, hence relating to A.

? Surface tension: Causes phenomena like capillary action where liquid defies gravity in narrow spaces, hence relating to D.

Hence, Option (1) is the right answer.

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2. Match the following chemical substances with their environmental impacts:

1. Substance Impact

I. Chloroform A. Ozone layer depletion

II. Freons B. Toxicity and potential carcinogenic effects

III. Iodoform C. Global warming potential

IV. DDT D. Bioaccumulation and ecological disruption

Choose the correct answer from the options given below:

(1) I-B, II-A, III-C, IV-D

(2) I-C, II-D, III-A, IV-B

(3) I-B, II-C, III-A, IV-D

(4) I-A, II-C, III-B, IV-D

Answer Key: 1

Solution:

? Chloroform: Known for its toxic and potentially carcinogenic effects when exposed to living organisms, hence relating to B.

? Freons: CFCs like Freons are notorious for their role in ozone depletion, hence relating to A.

? Iodoform: Used less commonly but can be a contributing factor to local environmental toxicity, hence relating to C.

? DDT: Known for its persistence in the environment and potential to bioaccumulate, leading to ecological disruption, hence relating to D.

Hence, Option (1) is the right answer.

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3. Match the following mathematical terms with their corresponding properties or relationships:

1. Term Property

I. Arithmetic Progression A. The sum of any two terms equidistant from the center equals the sum of the first and last term

II. Geometric Progression B. The square of the mean is equal to the product of the terms

III. Relation between A.M. and G.M. C. The product of terms in the sequence remains constant

IV. Insertion of means D. The mean is always greater than or equal to the geometric mean

Choose the correct answer from the options given below:

(1) I-A, II-C, III-D, IV-B

(2) I-D, II-A, III-B, IV-C

(3) I-C, II-D, III-B, IV-A

(4) I-B, II-C, III-A, IV-D

Answer Key: 1

Solution:

? Arithmetic Progression: The property of symmetry in AP where the sum of terms equidistant from the middle is constant, hence relating to A.

? Geometric Progression: In a GP, the ratio between successive terms is constant, which implies the product of terms equidistant in the series is constant, hence relating to C.

? Relation between A.M. and G.M.: This mathematical inequality states that the arithmetic mean of two non-negative numbers is greater than or equal to their geometric mean, hence relating to D.

? Insertion of means: When means are inserted between two numbers, the relationship between arithmetic and geometric means can be observed, hence relating to B.

Hence, Option (1) is the right answer.